

Programs on Decision Making and Knowledge Representation-2

Example 1 Family Relationships

Problem: Define family relations and answer grandparent queries.



```
p1 - Notepad
File Edit Format View Help
% Facts
parent(john, mary).
parent(mary, susan).
parent(john, mike).

% Rules
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).

% Query
% ?- grandparent(john, susan).
|
```

Output:



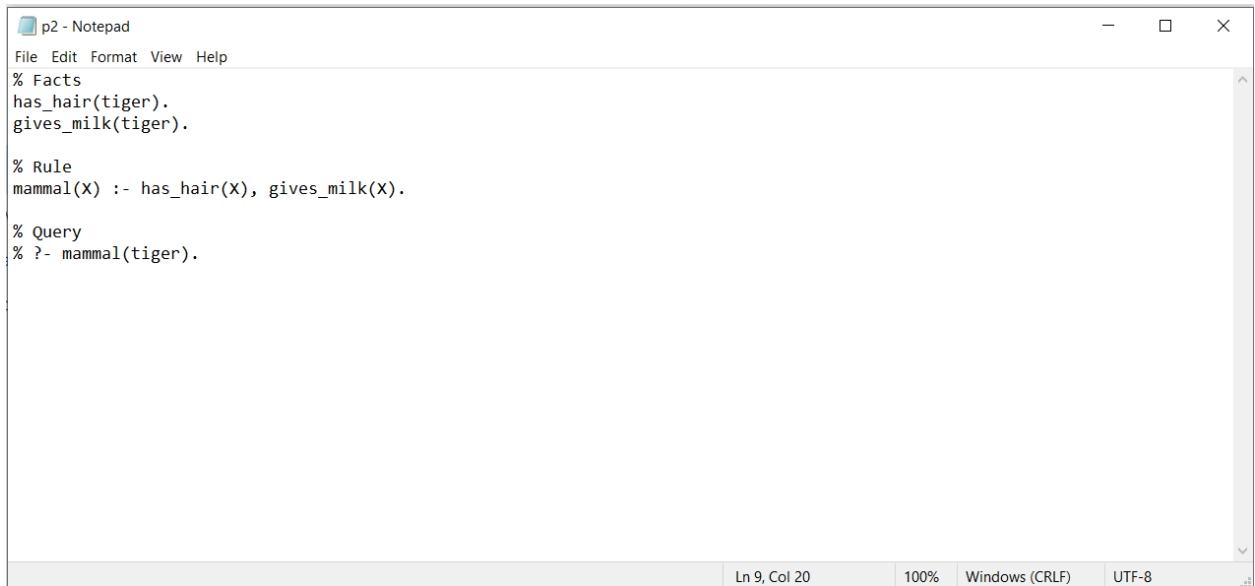
```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

4 ?- consult(['c:/Users/HDC0422167/Downloads/p1.pl']).
true.
```

2: Animal Classification

Problem: Identify if an animal is a mammal based on features.



```
p2 - Notepad
File Edit Format View Help
% Facts
has_hair(tiger).
gives_milk(tiger).

% Rule
mammal(X) :- has_hair(X), gives_milk(X).

% Query
% ?- mammal(tiger).
```

Ln 9, Col 20 100% Windows (CRLF) UTF-8

Output:

```
4 ?-
|   consult(['c:/Users/HDC0422167/Downloads/p2.pl']).
true.
```

3: Simple Arithmetic Knowledge Base

Problem: Implement simple addition using PROLOG.

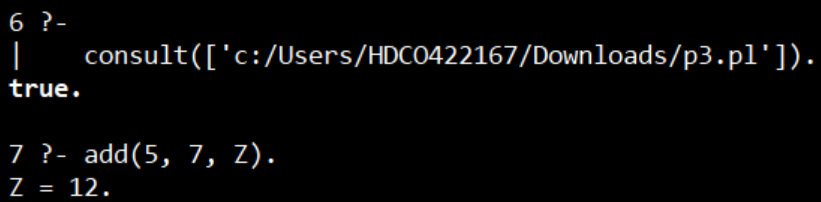


The screenshot shows a Notepad window with the title 'p3 - Notepad'. The menu bar includes 'File', 'Edit', 'Format', 'View', and 'Help'. The text content is as follows:

```
% Rule
add(X, Y, Z) :- Z is X + Y.
```

The status bar at the bottom indicates 'Ln 3, Col 1', '100%', 'Windows (CRLF)', and 'UTF-8'.

Output:



The screenshot shows the output of a Prolog interpreter. The text is as follows:

```
6 ?-
|   consult(['c:/Users/HDC0422167/Downloads/p3.pl']).
true.

7 ?- add(5, 7, Z).
Z = 12.
```